

```
Last login: Thu Jun 18 21:12:27 on ttys001
stephenrajva@Stephens-MacBook-Air-4 ~ % mkdir -p ~/Downloads/SST_Stimulator/MiniMD/Mesh
stephenrajva@Stephens-MacBook-Air-4 ~ % cp "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/mesh3D.py"
" \
~/Downloads/SST_Stimulator/MiniMD/Mesh/minimd_mesh16.py
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/SST_Stimulator/MiniMD/Mesh
stephenrajva@Stephens-MacBook-Air-4 Mesh % ls -l
total 8
-rw-r--r--  1 stephenrajva  staff   2872 19 Jun 12:31 minimd_mesh16.py
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "shape" minimd_mesh16.py
topo.shape = "2x2x4"
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "size" minimd_mesh16.py
"size": 16,
ep = EmberMPIJob(job["start"], job["size"])
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "filepath" minimd_mesh16.py
"filepath": "minimd_mesh_stats.csv",
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd ~/Downloads/SST_Stimulator/MiniMD/Mesh
stephenrajva@Stephens-MacBook-Air-4 Mesh % rm -f minimd_mesh_stats.csv
stephenrajva@Stephens-MacBook-Air-4 Mesh % sst minimd_mesh16.py
MiniMD Pattern Simulation Configured:
- Topology: 3D Mesh 2x2x4
- Hosts per router: 1
- Total compute nodes: 16
- Total ranks: 16
- Jobs configured: 1 job, 16 total ranks
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162340841053:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
162340841053:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
162340841053:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
162340841053:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
162341418675:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
162341418675:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
162341418675:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
162341418675:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
162343785364:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
162343785364:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
162343785364:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
162343785364:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 162344.182 us, loop 325, bufLen 20510, latency 249.760 us. bandwidth 0.082119 GB/s
162344362730:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162344362730:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
162344362730:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
162344362730:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
318234279315:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318234279315:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318234279315:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318234279315:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
318234801859:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318234801859:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318234801859:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318234801859:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318236008941:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318236008941:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318236008941:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318236008941:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155892.121 us, loop 325, bufLen 9000, latency 239.834 us. bandwidth 0.037526 GB/s
318236483589:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318236483589:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318236483589:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318236483589:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480575644936:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480575644936:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480575644936:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480575644936:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
480576222558:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480576222558:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480576222558:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480576222558:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480578589247:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480578589247:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
480578589247:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480578589247:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 162342.683 us, loop 325, bufLen 20510, latency 249.758 us. bandwidth 0.082120 GB/s
480579166613:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480579166613:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480579166613:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480579166613:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
AllPingPong: total time 155892.121 us, loop 325, bufLen 9000, latency 239.834 us. bandwidth 0.037526 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2458.075 us
Barrier: ranks 16, loop 15, latency 1399.178 us
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Simulation is complete, simulated time: 755.792 ms
stephenrajva@Stephens-MacBook-Air-4 Mesh % ls -lh *.csv
-rw-r--r--@ 1 stephenrajva  staff   48K 19 Jun 12:33 minimd_mesh_stats.csv
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "send_bit_count" minimd_mesh_stats.csv | head
rtr_0x0x0, send_bit_count, port0, Accumulator, 755792438438, 0, 22720, 9264640, 56, 360, 424
rtr_0x0x0, send_bit_count, port2, Accumulator, 755792438438, 0, 22720, 9264640, 56, 360, 424
rtr_0x0x0, send_bit_count, port4, Accumulator, 755792438438, 0, 154178640, 2473227262080, 12462, 104, 16384
rtr_0x0x0, send_bit_count, port6, Accumulator, 755792438438, 0, 154178640, 2473227262080, 12462, 104, 16384
nic0:rtrLink:networkIF, send_bit_count, , Accumulator, 755792438438, 0, 154178640, 2473227262080, 12462, 104, 16384
rtr_1x0x0, send_bit_count, port1, Accumulator, 755792438438, 0, 45440, 18529280, 112, 360, 424
rtr_1x0x0, send_bit_count, port2, Accumulator, 755792438438, 0, 22720, 9264640, 56, 360, 424
rtr_1x0x0, send_bit_count, port4, Accumulator, 755792438438, 0, 154155920, 2473217997440, 12406, 104, 16384
rtr_1x0x0, send_bit_count, port6, Accumulator, 755792438438, 0, 154201360, 2473236526720, 12518, 104, 16384
nic1:rtrLink:networkIF, send_bit_count, , Accumulator, 755792438438, 0, 154201360, 2473236526720, 12518, 104, 16384
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "send_packet_count" minimd_mesh_stats.csv | head
rtr_0x0x0, send_packet_count, port0, Accumulator, 755792438438, 0, 56, 56, 56, 1, 1
rtr_0x0x0, send_packet_count, port2, Accumulator, 755792438438, 0, 56, 56, 56, 1, 1
rtr_0x0x0, send_packet_count, port4, Accumulator, 755792438438, 0, 12462, 12462, 12462, 1, 1
rtr_0x0x0, send_packet_count, port6, Accumulator, 755792438438, 0, 12462, 12462, 12462, 1, 1
rtr_1x0x0, send_packet_count, port1, Accumulator, 755792438438, 0, 112, 112, 112, 1, 1
rtr_1x0x0, send_packet_count, port2, Accumulator, 755792438438, 0, 56, 56, 56, 1, 1
rtr_1x0x0, send_packet_count, port4, Accumulator, 755792438438, 0, 12406, 12406, 12406, 1, 1
rtr_1x0x0, send_packet_count, port6, Accumulator, 755792438438, 0, 12518, 12518, 12518, 1, 1
rtr_0x1x0, send_packet_count, port0, Accumulator, 755792438438, 0, 56, 56, 56, 1, 1
rtr_0x1x0, send_packet_count, port3, Accumulator, 755792438438, 0, 112, 112, 112, 1, 1
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "output_port_stalls" minimd_mesh_stats.csv | head
rtr_0x0x0, output_port_stalls, port0, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_0x0x0, output_port_stalls, port2, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_0x0x0, output_port_stalls, port4, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_0x0x0, output_port_stalls, port6, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
nic0:rtrLink:networkIF, output_port_stalls, , Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_1x0x0, output_port_stalls, port1, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_1x0x0, output_port_stalls, port2, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_1x0x0, output_port_stalls, port4, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
rtr_1x0x0, output_port_stalls, port6, Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
nic1:rtrLink:networkIF, output_port_stalls, , Accumulator, 755792438438, 0, 0, 0, 0, 0, 0
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "idle_time" minimd_mesh_stats.csv | head
rtr_0x0x0, idle_time, port0, Accumulator, 755792438438, 0, 755792161174, 5948676043217932894, 57, 3262040, 638925365030
rtr_0x0x0, idle_time, port2, Accumulator, 755792438438, 0, 755792161174, 6114029836295935854, 57, 3131975, 638925494428
rtr_0x0x0, idle_time, port4, Accumulator, 755792438438, 0, 754185485610, 18352588806626779956, 3363, 930469, 2457704137
rtr_0x0x0, idle_time, port6, Accumulator, 755792438438, 0, 754185485610, 486695755104482794, 9216, 69372, 2457627537
nic0:rtrLink:networkIF, idle_time, , Accumulator, 755792438438, 0, 434415746981, 10778403884398194001, 3362, 124668, 2456441375
rtr_1x0x0, idle_time, port1, Accumulator, 755792438438, 0, 755791883910, 3976569935570844310, 113, 1873607, 638924165097
rtr_1x0x0, idle_time, port2, Accumulator, 755792438438, 0, 755792161174, 7066071826442600092, 57, 2387603, 638926239467
rtr_1x0x0, idle_time, port4, Accumulator, 755792438438, 0, 754185762874, 124774671595518598, 3307, 1207273, 2458173810
rtr_1x0x0, idle_time, port6, Accumulator, 755792438438, 0, 754185208346, 18429288023065019308, 9272, 72164, 2455178313
nic1:rtrLink:networkIF, idle_time, , Accumulator, 755792438438, 0, 434355936251, 10282588650993873785, 3418, 124668, 2454316980
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "packet_latency" minimd_mesh_stats.csv | head
nic0:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 9587401, 10458560421, 12462, 143, 1863
nic1:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 9598847, 10460900769, 12518, 143, 1863
nic2:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 9602296, 10462520520, 12518, 142, 1863
nic3:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 9605720, 10464556350, 12518, 142, 1863
nic4:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 6778681, 5927460133, 12518, 204, 1741
nic5:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 6785539, 5930686551, 12518, 204, 1741
nic6:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 6782110, 5929917422, 12518, 143, 1741
nic7:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 6763778, 5923917888, 12462, 143, 1741
nic8:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 8527910, 8042752710, 12406, 204, 1854
nic9:rtrLink:networkIF, packet_latency, , Accumulator, 755792438438, 0, 8521037, 8039097873, 12406, 204, 1854
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd ~/Downloads/SST_Stimulator/MiniMD/Mesh
stephenrajva@Stephens-MacBook-Air-4 Mesh % cp minimd_mesh16.py minimd_mesh64.py
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "shape" minimd_mesh64.py
topo.shape = "2x2x4"
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "size" minimd_mesh64.py
"size": 16,
ep = EmberMPIJob(job["start"], job["size"])
stephenrajva@Stephens-MacBook-Air-4 Mesh % nano minimd_mesh64.py
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "shape" minimd_mesh64.py
topo.shape = "4x4x4"
stephenrajva@Stephens-MacBook-Air-4 Mesh %
grep "size" minimd_mesh64.py
"size": 64,
ep = EmberMPIJob(job["start"], job["size"])
stephenrajva@Stephens-MacBook-Air-4 Mesh % grep "filepath" minimd_mesh64.py
"filepath": "minimd_mesh_stats64.csv",
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd ~/Downloads/SST_Stimulator/MiniMD/Mesh
stephenrajva@Stephens-MacBook-Air-4 Mesh % rm -f minimd_mesh_stats64.csv
stephenrajva@Stephens-MacBook-Air-4 Mesh % sst minimd_mesh64.py
MiniMD Pattern Simulation Configured:
- Topology: 3D Mesh 2x2x4
- Hosts per router: 1
- Total compute nodes: 16
- Total ranks: 64
- Jobs configured: 1 job, 64 total ranks
180000:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
180000:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
180000:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
180000:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
180000:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
180000:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
180000:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
180000:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
180000:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
180000:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
180000:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
180000:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64
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[illegible]

[illegible]

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480575644936:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
480575644936:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
480575644936:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
480575644936:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
480575644936:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
480575644936:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
480575644936:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
480575644936:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
480576222558:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=64
480576222558:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=64
480576222558:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=64
480576222558:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=64
480576222558:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=64
480576222558:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
480576222558:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
480576222558:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
480576222558:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
480576222558:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
480576222558:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
480576222558:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
480576222558:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
480576222558:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
480576222558:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
480576222558:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
480578589247:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
480578589247:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
480578589247:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
480578589247:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
480578589247:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
480578589247:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
480578589247:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
480578589247:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
480578589247:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
480578589247:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
480578589247:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
480578589247:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
480578589247:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
480578589247:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
480578589247:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
480578589247:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
AllPingPong: total time 162342.683 us, loop 325, buflen 20510, latency 249.758 us. bandwidth 0.082120 GB/s
480579166613:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
480579166613:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
480579166613:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
480579166613:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
480579166613:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
480579166613:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
480579166613:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
480579166613:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
480579166613:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
480579166613:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
480579166613:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
480579166613:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
480579166613:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
480579166613:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64
480579166613:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
480579166613:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
AllPingPong: total time 155892.121 us, loop 325, buflen 9000, latency 239.834 us. bandwidth 0.037526 GB/s
Allreduce: ranks 64, loop 40, 1 double(s), latency 2462.205 us
Barrier: ranks 64, loop 15, latency 1403.354 us
Simulation is complete, simulated time: 756.026 ms
stephenrajva@Stephens-MacBook-Air-4 Mesh % ls -lh minimd_mesh_stats64.csv
-rw-r--r-- 1 stephenrajva staff 222K 19 Jun 12:39 minimd_mesh_stats64.csv
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd ~/Downloads/SST_Stimulator/MiniMD/Mesh
stephenrajva@Stephens-MacBook-Air-4 Mesh % nano extract_mesh_metrics.py
stephenrajva@Stephens-MacBook-Air-4 Mesh % python3 extract_mesh_metrics.py
Traceback (most recent call last):
  File "/Users/stephenrajva/Downloads/SST_Stimulator/MiniMD/Mesh/extract_mesh_metrics.py", line 24, in <module>
    avg_latency = latency[9].mean() if len(latency) else 0
                                ~~~~~^
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/util/_decorators.py", line 336, in wrapper
    return func(*args, **kwargs)
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/series.py", line 8113, in mean
    return NDFrame.mean(
           ~~~~~^
        self, axis=axis, skipna=skipna, numeric_only=numeric_only, **kwargs
        ~~~~~^
    )
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/generic.py", line 11831, in mean
    return self._stat_function(
           ~~~~~^
        "mean", nanops.nanmean, axis, skipna, numeric_only, **kwargs
        ~~~~~^
    )
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/generic.py", line 11785, in _stat_function
    return self._reduce(
           ~~~~~^
        func, name=name, axis=axis, skipna=skipna, numeric_only=numeric_only
        ~~~~~^
    )
  File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/series.py", line 7480, in _reduce
    result = delegate._reduce(name, skipna=skipna, **kwargs)

```

```
File "/opt/homebrew/lib/python3.14/site-packages/pandas/core/arrays/string_.py", line 969, in _reduce
    raise TypeError(f"Cannot perform reduction '{name}' with string dtype")
```

```
TypeError: Cannot perform reduction 'mean' with string dtype
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % nano extract_mesh_metrics.py
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % python3 extract_mesh_metrics.py
```

```
File "/Users/stephenrajva/Downloads/SST_Stimulator/MiniMD/Mesh/extract_mesh_metrics.py", line 1
```

```
import pandas as pd
```

```
AAAAAAA
```

```
SyntaxError: invalid syntax. Did you mean 'import'?
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % nano extract_mesh_metrics.py
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % python3 extract_mesh_metrics.py
```

```
Processing minimd_mesh_stats.csv...
```

```
Processing minimd_mesh_stats64.csv...
```

```
===== MESH COMPARISON =====
```

	Topology	SendBits	Packets	Stalls	AvgLatency	AvgIdleTime	Throughput
0	Mesh16	9867296640	597952	0	181.000000	6.860887e+11	5.451545e+07
1	Mesh64	39473230720	2401440	0	194.453125	6.970467e+11	2.029961e+08

```
Saved: mesh_summary.csv
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % cd ~/Downloads/SST_Stimulator/MiniMD/Mesh
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % nano plot_mesh_scaling.py
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % python3 plot_mesh_scaling.py
```

```
Created:
```

```
mesh_latency.png
```

```
mesh_packets.png
```

```
mesh_throughput.png
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh %
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh % mkdir -p ~/Downloads/SST_Stimulator/MiniMD/Torus
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh %
```

```
cp "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/topoTorus/topoTorus.py" \
```

```
~/Downloads/SST_Stimulator/MiniMD/Torus/minimd_torus16.py
```

```
cp: /Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/topoTorus/topoTorus.py: No such file or directory
```

```
stephenrajva@Stephens-MacBook-Air-4 Mesh %
```

```
cd ~/Downloads/SST_Stimulator/MiniMD/Torus
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -l
```

```
total 0
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % cp "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/torus/torus.py" \
```

```
"/Users/stephenrajva/Downloads/SST_Stimulator/MiniMD/Torus/minimd_torus16.py"
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % cd "/Users/stephenrajva/Downloads/SST_Stimulator/MiniMD/Torus"
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -l
```

```
total 8
```

```
-rw-r--r--  1 stephenrajva  staff   2876  19 Jun 12:51 minimd_torus16.py
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "shape" minimd_torus16.py
```

```
topo.shape = "2x2x4"
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "size" minimd_torus16.py
```

```
"size": 16,
```

```
ep = EmberMPIJob(job["start"], job["size"])
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "filepath" minimd_torus16.py
```

```
"filepath": "minimd_torus_stats.csv",
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniMD/Torus
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % rm -f minimd_torus_stats.csv
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % sst minimd_torus16.py
```

```
MiniMD Pattern Simulation Configured:
```

```
- Topology: 3D torus 2x2x4
```

```
- Hosts per router: 1
```

```
- Total compute nodes: 16
```

```
- Total ranks: 16
```

```
- Jobs configured: 1 job, 16 total ranks
```

```
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
```

```
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
```

```
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
```

```
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
```

```
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
```

```
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
```

```
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
```

```
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
```

```
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
```

```
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
```

```
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
```

```
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
```

```
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
```

```
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
```

```
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
```

```
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
```

```
162380668621:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
```

```
162380668621:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
```

```
162380668621:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
```

```
162380668621:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
```

```
162381118179:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
```

```
162381118179:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
```

```
162381118179:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
```

```
162381118179:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
```

```
162383592383:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
```

```
162383592383:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
```

```
162383592383:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
```

```
162383592383:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
```

```
AllPingPong: total time 162383.861 us, loop 325, bufLen 20510, latency 249.821 us. bandwidth 0.082099 GB/s
```

```
162384041557:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
```

```
162384041557:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
```

```
162384041557:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
```

```

162384041557:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
318232182598:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318232182598:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318232182598:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318232182598:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
318232699806:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318232699806:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318232699806:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318232699806:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318233762149:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318233762149:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318233762149:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318233762149:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155850.238 us, loop 325, bufLen 9000, latency 239.770 us. bandwidth 0.037536 GB/s
318234279357:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318234279357:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318234279357:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318234279357:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480613738971:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480613738971:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480613738971:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480613738971:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
480614188529:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480614188529:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480614188529:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480614188529:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480616662733:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480616662733:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
480616662733:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480616662733:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 162382.832 us, loop 325, bufLen 20510, latency 249.820 us. bandwidth 0.082099 GB/s
480617111907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480617111907:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480617111907:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480617111907:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
AllPingPong: total time 155850.238 us, loop 325, bufLen 9000, latency 239.770 us. bandwidth 0.037536 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2458.205 us
Barrier: ranks 16, loop 15, latency 1399.311 us
Simulation is complete, simulated time: 755.796 ms
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -lh *.csv
-rw-r--r-- 1 stephenrajva staff 66K 19 Jun 12:52 minimd_torus_stats.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniMD/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % cp minimd_torus16.py minimd_torus64.py
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "shape" minimd_torus64.py
topo.shape = "2x2x4"
stephenrajva@Stephens-MacBook-Air-4 Torus %
grep "size" minimd_torus64.py
"size": 16,
ep = EmberMPIJob(job["start"], job["size"])
stephenrajva@Stephens-MacBook-Air-4 Torus %
grep "filepath" minimd_torus64.py
"filepath": "minimd_torus_stats.csv",
stephenrajva@Stephens-MacBook-Air-4 Torus % nano minimd_torus64.py
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "shape" minimd_torus64.py
topo.shape = "4x4x4"
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "size" minimd_torus64.py
"size": 64,
ep = EmberMPIJob(job["start"], job["size"])
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "filepath" minimd_torus64.py
"filepath": "minimd_torus_stats64.csv",
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniMD/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus %
rm -f minimd_torus_stats64.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % sst minimd_torus64.py
MiniMD Pattern Simulation Configured:
- Topology: 3D torus 2x2x4
- Hosts per router: 1
- Total compute nodes: 16
- Total ranks: 64
- Jobs configured: 1 job, 64 total ranks
180000:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
180000:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
180000:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
180000:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
180000:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
180000:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
180000:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
180000:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
180000:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
180000:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
180000:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
180000:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64
180000:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=64
180000:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=64
180000:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=64
180000:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=64
180000:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
180000:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
180000:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
180000:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
180000:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
180000:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
180000:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
180000:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
180000:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64

```

[illegible]

[illegible]

```

480614188529:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
480614188529:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
480614188529:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
480614188529:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
480614188529:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
480614188529:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
480614188529:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
480614188529:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
480614188529:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
480614188529:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
480614188529:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
480616662733:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
480616662733:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
480616662733:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
480616662733:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
480616662733:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
480616662733:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
480616662733:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
480616662733:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
480616662733:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
480616662733:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
480616662733:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
480616662733:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
480616662733:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
480616662733:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
480616662733:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
480616662733:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
AllPingPong: total time 162382.832 us, loop 325, bufLen 20510, latency 249.820 us. bandwidth 0.082099 GB/s
480617111907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
480617111907:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
480617111907:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
480617111907:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
480617111907:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
480617111907:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
480617111907:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
480617111907:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
480617111907:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
480617111907:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
480617111907:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
480617111907:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
480617111907:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
480617111907:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64
480617111907:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
480617111907:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
AllPingPong: total time 155850.238 us, loop 325, bufLen 9000, latency 239.770 us. bandwidth 0.037536 GB/s
Allreduce: ranks 64, loop 40, 1 double(s), latency 2462.436 us
Barrier: ranks 64, loop 15, latency 1403.584 us
Simulation is complete, simulated time: 756.035 ms
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -lh *.csv
-rw-r--r-- 1 stephenrajva staff 66K 19 Jun 12:52 minimd_torus_stats.csv
-rw-r--r-- 1 stephenrajva staff 263K 19 Jun 12:56 minimd_torus_stats64.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniMD/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % nano extract_torus_metrics.py
stephenrajva@Stephens-MacBook-Air-4 Torus % python3 extract_torus_metrics.py

Processing minimd_torus_stats.csv...

Processing minimd_torus_stats64.csv...

===== TORUS COMPARISON =====
Topology SendBits Packets Stalls AvgLatency AvgIdleTime Throughput
0 Torus16 9867296640 597952 0 199.2500 7.078849e+11 4.952219e+07
1 Torus64 39472503680 2399648 0 213.6875 7.081248e+11 1.847207e+08

Saved: torus_summary.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniMD/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % nano plot_torus_scaling.py
stephenrajva@Stephens-MacBook-Air-4 Torus % python3 plot_torus_scaling.py
Created:
torus_latency.png
torus_packets.png
torus_throughput.png
stephenrajva@Stephens-MacBook-Air-4 Torus % mkdir -p ~/Downloads/SST_Stimulator/MiniMD/Comparison
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniMD/Comparison
stephenrajva@Stephens-MacBook-Air-4 Comparison %
cp ~/Downloads/SST_Stimulator/MiniMD/Mesh/mesh_summary.csv .
stephenrajva@Stephens-MacBook-Air-4 Comparison % cp ~/Downloads/SST_Stimulator/MiniMD/Torus/torus_summary.csv .
stephenrajva@Stephens-MacBook-Air-4 Comparison % ls -lh
total 16
-rw-r--r-- 1 stephenrajva staff 212B 19 Jun 13:01 mesh_summary.csv
-rw-r--r-- 1 stephenrajva staff 213B 19 Jun 13:01 torus_summary.csv
stephenrajva@Stephens-MacBook-Air-4 Comparison % nano combine_minimd_results.py
stephenrajva@Stephens-MacBook-Air-4 Comparison % python3 combine_minimd_results.py

===== COMBINED MINIMD DATA =====
Topology SendBits Packets Stalls AvgLatency AvgIdleTime Throughput Family
0 Mesh16 9867296640 597952 0 181.000000 6.860887e+11 5.451545e+07 Mesh
1 Mesh64 39473230720 2401440 0 194.453125 6.970467e+11 2.029961e+08 Mesh
2 Torus16 9867296640 597952 0 199.250000 7.078849e+11 4.952219e+07 Torus
3 Torus64 39472503680 2399648 0 213.687500 7.081248e+11 1.847207e+08 Torus

Saved: combined_minimd_comparison.csv
stephenrajva@Stephens-MacBook-Air-4 Comparison % nano plot_minimd_comparison.py
stephenrajva@Stephens-MacBook-Air-4 Comparison % python3 plot_minimd_comparison.py
Created:

```